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ENTERPRISE ARCHITECT — HIGH-SCALE SYSTEMS & DATA-INTENSIVE PLATFORMS

Real-Time Transaction Processing • Data Pipeline Architecture • ML/Analytics Infrastructure • System Modernization

Enterprise architect with 18+ years designing mission-critical, data-intensive systems at scale. Deep expertise in high-volume transaction platforms (1B+ transactions/month), real-time data pipelines, and complex system integrations across distributed architectures. Recent focus: leading enterprise-scale migrations and modernization initiatives with strong emphasis on data governance, observability, and intelligent automation. Proven ability to architect resilient systems that process massive data volumes while maintaining auditability, traceability, and operational excellence.

CORE CAPABILITIES

System Architecture: High-scale distributed systems, real-time data processing, event-driven architecture, API/service design, integration patterns, resiliency engineering

Data & Analytics: Data pipeline architecture, streaming data systems, analytics infrastructure, data lineage/governance, metrics-driven insights, ML pipeline integration

Enterprise Delivery: Target-state design, modernization roadmaps, cross-functional alignment (product, engineering, ops, risk), stakeholder management, production readiness

Technology Stack: Python, Java, Spring Boot, SQL/Oracle, Kafka-style streaming, REST/API design, observability tools, data analysis (Jupyter, Tableau, Alteryx)

Domain Expertise: Financial services, payments systems (ACH/Wire/Real-time), fraud/risk integration, regulatory compliance, 24/7 operational systems

PROFESSIONAL EXPERIENCE

BANK OF AMERICA — Charlotte, NC

Senior Architect, Enterprise Technology | Jun 2018 – Present

Enterprise Platform Migration & Modernization (2024–2026)

Leading architecture and delivery for enterprise-scale platform migration supporting 26M customers, 6M active digital users, processing ~1M transactions/month—on track for 2026 completion enabling ~\$100M total cost optimization.

- **Designed end-to-end data flow architecture** across digital channels → transaction processing → fraud/risk decisioning → settlement systems, defining service contracts, data models, sequencing, error handling, and observability requirements
- **Architected complex data migration strategy** for customer data, transaction history, scheduled events, and in-flight processing during zero-downtime cutover windows
- **Built analytical frameworks** to validate data integrity, reconcile discrepancies, and provide audit evidence across 20+ data domains using Python/SQL analysis pipelines
- **Drove delivery across distributed engineering teams** implementing microservices (Spring Boot/Oracle) with production-grade NFRs including SLAs, monitoring, and operational runbooks
- **Established data governance practices** for transaction traceability, regulatory reporting, and downstream analytics consumption

Real-Time Processing & Intelligent Systems (2022–2023)

Designed architecture enhancements enabling real-time settlement and decisioning for high-velocity transaction workflows.

- **Architected real-time data pipeline** with sub-second latency requirements, implementing idempotency, retry strategies, failure isolation, and distributed tracing
- **Integrated ML-based fraud detection** into transaction flows, designing fallback modes and degraded operations to maintain availability during model/service outages
- **Implemented observability framework** for real-time monitoring, anomaly detection, and automated alerting across transaction lifecycle

Data Platform & Analytics Infrastructure (2018–2022)

Enhanced mainframe platforms processing 1B+ transactions/month while modernizing data accessibility and analytics capabilities.

- **Built data extraction and analysis pipelines** supporting regulatory validation and audit requirements, reducing evidence delivery time by 60% through standardized workflows

- **Developed analytical models** using Jupyter Notebooks and Python to identify patterns in transaction failures, processing bottlenecks, and system behavior—informing architecture improvements
 - **Created metrics dashboards** (Tableau) providing real-time visibility into system health, transaction volumes, success rates, and operational KPIs
 - **Led modernization initiatives** to expose mainframe data through APIs, enabling downstream analytics and ML model consumption
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PENFED CREDIT UNION — McLean, VA

Senior Solution Architect | Jul 2017 – May 2018

- **Designed and delivered ML model integration** migrating credit decisioning from legacy Equifax Pinnacle to FICO 9, including data pipeline modifications and model validation framework
 - **Architected rapid product launch** for American Express credit card platform in 3 months through pragmatic technical decisions and parallel workstream coordination
 - **Led development teams** delivering complex customer-facing workflows with data integration across multiple systems
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TATA CONSULTANCY SERVICES — Multiple Locations

Business Relationship Manager | Apr 2013 – Jul 2017

- **Managed global delivery organization** of 150+ engineers across US/Mexico/India supporting Bank of America enterprise platforms
 - **Scaled team 3× (50→150)** while maintaining >35% profit margin and meeting SLAs across development, testing, production support
 - **Drove operational excellence** through metrics-driven management, automated reporting, and continuous improvement initiatives
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BANK OF AMERICA — Charlotte, NC

Senior Technical Lead, Transaction Processing Platforms | Nov 2004 – Mar 2013

- **Designed and delivered large-scale system integrations** for bank acquisitions (LaSalle, Countrywide, U.S. Trust) processing millions of customer accounts and transaction histories

- **Built high-availability transaction authorization systems** supporting new digital channels with strict latency and reliability requirements
 - **Established engineering practices** for testing, deployment automation, and production stability across distributed teams
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TOYOTA FINANCIAL SERVICES — Torrance, CA

Technical Analyst, Commercial Systems | May 2003 – Oct 2004

- **Automated data-intensive workflows** replacing manual spreadsheet processes with SQL/DB2-based pipelines, reducing processing time by 80%
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TECHNICAL PROJECTS & CONTINUOUS LEARNING

Applied Data Science Program (2021) — 12-week intensive program covering deep learning, time series analysis, recommendation systems, and ML case studies with hands-on implementation

Current Focus (2026): Exploring applications of large language models and generative AI to enterprise data problems—investigating RAG architectures, prompt engineering patterns, and LLM integration strategies for document analysis and intelligent automation

EDUCATION

B.E., Computer Science & Engineering — College of Engineering, Anna University, Chennai, India

KEY DIFFERENTIATORS

- **Scale:** Architected systems processing 1B+ transactions/month with 99.99%+ availability requirements
- **Data Intensity:** Deep experience with high-volume data pipelines, real-time streaming, analytics infrastructure
- **Modernization:** Led transitions from legacy mainframe to modern microservices while maintaining operational continuity

- **Business Impact:** Consistently delivered outcomes tied to measurable business value (\$100M+ cost optimization, 60% efficiency gains)
- **Cross-functional Leadership:** Proven ability to align technical architecture with product, operations, risk, and compliance stakeholders